

Full Length Article

Sentiment analysis on Twitter data towards climate action

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ABSTRACT

Understanding the progress of the Sustainable Development Goals (SDGs) proposed by the United Nations (UN) is important, but difficult. In particular, policymakers would need to understand the sentiment within the public regarding challenges associated with climate change. With this in mind and the rise of social media, this work focuses on the task of uncovering the sentiment of Twitter users concerning climate-related issues. This is done by applying modern natural-language-processing (NLP) methods, *i.e.* VADER, TextBlob, and BERT, to estimate the sentiment of a gathered dataset based on climate-change keywords. A transfer-learning-based model applied to a pre-trained BERT model for embedding and tokenizing with logistic regression for sentiment classification outperformed the rule-based methods VADER and TextBlob; based on our analysis, the proposed approach led to the highest accuracy: 69%. The collected data contained significant noise, especially from the keyword ‘energy’. Consequently, using more specific keywords would improve the results. The use of other methods, like BERTweet, would also increase the accuracy of the model. The overall sentiment in the analyzed data was positive. The distribution of the positive, neutral, and negative sentiments was very similar in the different SDGs.

1. Introduction

During the last two decades, social media has become a large part of people’s daily life. It is a communication channel where ideas and thoughts are shared and as such, it is a tool that shapes the future generation. Social networks open up the possibility for people everywhere to connect and make a difference and through that, it has the power to push instant change. Both industry and academia seek to understand why people act and to what extent they act as a group. Social media is an ideal platform to study these aspects thanks to the large amount of data that is available.

A famous, highly controversial, example is Cambridge Analytica which used data from social media, especially Facebook, to perform data analysis for political advertising [1]. Twitter, with its short text messages in the form of tweets posted by the users, can be used in a similar way to analyze the attitude of the speaker on a certain topic and by aggregation, identify the public opinion on that topic. Natural language processing (NLP) can be used to mine and infer the sentiment of a given tweet, which in turn can be used for inferring the collective opinion about a topic. The complexity of the written language makes this a fairly complicated task and much effort has been spent on untangling this problem. In this sense, as argued by Vinuesa and Sirmacek, it is essential to use interpretable deep-learning models when they have implications for their SDGs. [2] There are many pre-processing techniques to aid the sentiment classification problem, such as removing stop words, lemming, stemming, and changing emojis to words. Next, sentiment classification can be based on statistical methods *e.g.* support-vector machine (SVM) and naive Bayes, as well as rule-based models such as VADER (Valence Aware Dictionary for sentiment Reasoning), TextBlob and Linguistic Inquiry and Word Count (LIWC). Deep-learning-based models that rely on training deep neural networks are, however, taking over the field; one example is the BERT model (Bidirectional Encoder Representations from Transformers). Using NLP analysis to assess public opinion regarding climate change is still underresearched. Climate change is more relevant than ever and much needs to be done in order to reach the goals stipulated in the Agenda for 2030 of the United Nations (UN) [3].

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Understanding the effect different narratives have on global change (behavioral pattern, engagement in movements, etc.) would be helpful, if not necessary, for achieving the global goals by 2030. To get a better understanding of the current sentiment surrounding climate change among the general public, an analysis of Twitter data could provide valuable insights. NLP has proven effective for understanding collective opinions and sentiments. One way of doing this is by using keywords relative to the Sustainable Development Goal (SDG) 7, 11, and 13 on affordable and clean energy, sustainable cities and communities. The project will therefore perform sentiment analysis on Twitter data related to these SDGs.

When it comes to previous work, studies have been made on analysis behavior and public opinion using Twitter data. The use of NLP techniques is common but an analysis of the methodology itself is more rare. The easy access to NLP methods, such as VADER and pre-trained BERT models, open up the opportunity for many scientists, independent of their background and scientific field, to use the methodology. The large language model (LLM) GPT is not part of the study, GPT is a different type of model than BERT (generative and encoding-decoding based) and therefore not the typical choice when it comes to classifications task such as sentiment analysis. In this work, we compare baseline methods in contrast to the rule-based methods used in the past and therefore uses BERT. This paper will show that the choice of method can be a decisive factor in the result.

The classification of the semantics in words with lexicon-based methods started with document classifications, using overall sentiment. Pang et al. (2002) [4] used traditional text-categorization techniques, Turney (2002) predicted the recommendation of movie reviews using the average semantic orientation of the phrases in review [5] and Dave et al. (2003) [6] developed a classification method based on heuristic methods. Moving on from documentary classification to sentence-based classification such as the work by Appel et al. (2016), [7] who used a hybrid lexicon-based method to predict the sentiments of sentences. Wilson et al. (2005) [8] provided a solution that determines the sentiment on a phrase level by first determining if the expression is neutral or polar and then disambiguating the polarity of the expression. On the document level, Moraes et al. (2013) [9] compared Support Vector Machines (SVM) and Artificial Neural Network (ANN) for sentiment analysis and Johnson and Zhang (2015) [10] proposed a method based on a two-view semi-supervised learning framework with convolutional neural networks (CNNs) for text embedding. The method learns embeddings of small text regions from unlabeled data and then the labeled data. At the sentence level within the area of deep-learning methods Guggilla et al. (2016) [11] proposed a supervised method based on CNN and long-short-term memory (LSTM) which used Word2Vec [12] and linguistic embeddings for claim classification. On aspect-level sentiment classification, Ruder et al. (2016) [13] created a hierarchical bidirectional LSTM that has the ability to leverage both intra- and inter-sentence relations. In 2017 Vaswani et al. published Attention is all that you need [14] and presented the Transformer that changed the field. Pipalia et al. (2020) [15] did a comparative analysis of different Transformer based architectures used in sentiment analysis and Neelakandan et al. (2022) [16] presented a Deep Learning Modified Neural Network (DLMNN) based proficient sentiment analysis technique on Twitter data.

Twitter has become a popular source and attracted more research interest in the last couple of years. The methods for sentiment analysis follow the traditional ones. Zimbra et al. (2018) [17] performed a review over the state-of-the-art methods in research of Twitter sentiment analysis. Within that there is research on sentiment analysis on tweets from Twitter that contains keywords that are related to climate change. However, to date, sentiment analyses focused much less that focus on the SDGs. Loureiro and Alló (2020) [18] conducted sentiment analysis on Twitter data collected over six months from Spain and U.K. The collected data contained one or more pre-defined keywords related to energy issues and climate change, they used a dictionary-based method where some examples of the semantic origination in the lexicon were anger, joy, fear, and surprise. They found that the overall sentiment in the two countries matched the general opinion about the climate in those countries. Kirilenko et al. (2014) [19] did a major study based on tweets collected over one year from all around the globe. They concluded that Twitter data contribute to communication research with its availability and through the concept of people as sensors. Maynard and Boncheva (2015) [20] developed an open source toolkit for sentiment analysis of climate change tweets with the aim to understand the engagement of the public with the respect to the topic of climate change and Reyes-Menedez et al. (2018) [21] conducted a topic-based sentiment analysis to understand public opinion surrounding the hashtag #WorldEnvironmentDay. There has been some research on artificial intelligence's (AI) role in helping archiving the Agenda 2030. Vinuesa et al. (2020) [22] published an article that looked into the role artificial intelligence has in achieving the SDGs and Gupta et al. (2021) [23] assessed whether artificial intelligence is an enabler or an inhibitor of sustainability at the indicator level. MacFeely [24] examined the opportunities and challenges big data has for compiling SDG indicators. Sánchez-Roncero et al. (2023) [25] used AI to develop ASDG, a model for automatically identifying the potential impact of scientific papers on the UN SDGs. Furthermore, recent work [26–28] has focused on AI regulation and its impact on societies.

2. Data collection

In this work, we first describe the process of data collection from tweets using keywords, and the pre-processing used. In the next section, the NLP methods used (VADER, TextBlob and BERT) are presented, and later the results obtained analyzed in the last section.

Two types of data have been used: the data used for training the logistic regression model and the data collected through the Twitter API for analysis. The Twitter API data was collected on a weekly basis, and the number of tweets collected could vary from week to week.

2.1. Keywords

Before downloading the tweets from Twitter it was necessary to decide which tweets to include. Since the goal of the paper is to analyze Twitter data related to SDGs, the tweets had to be related to them. There are many different ways to do this, for example following hashtags on Twitter or choosing tweets that contain keywords related to the topic. This paper will cover the last option.

Keywords related to each of the targets to each of the SDGs were chosen. The number of keywords is presented in the Table 1 and examples of keywords are presented in Fig. 1. A full list of the keywords can be found in the appendix.

2.2. Twitter API

The data is collected live with the Twitter API, from Twitter's public timeline. It is available as a free version and as a research version that allows the user to download tweets, location, and other features. The access to the Twitter API was done with the help of the open source package Tweepy, a package that helps automate the use of the Twitter API [29]. With this package, it is possible to easily define the keywords and pull tweets containing those keywords. It is possible to use both expressions, such as "Data we love", and single words like "Twitter", and pick multiple words to seek for tweets that contain both, like "Tweets, data". In this case both have been used. The Twitter API matches the keyword against the

Table 1
Keywords statistics.

SDG	Keywords
7	35
11	43
13	35
Tot	113

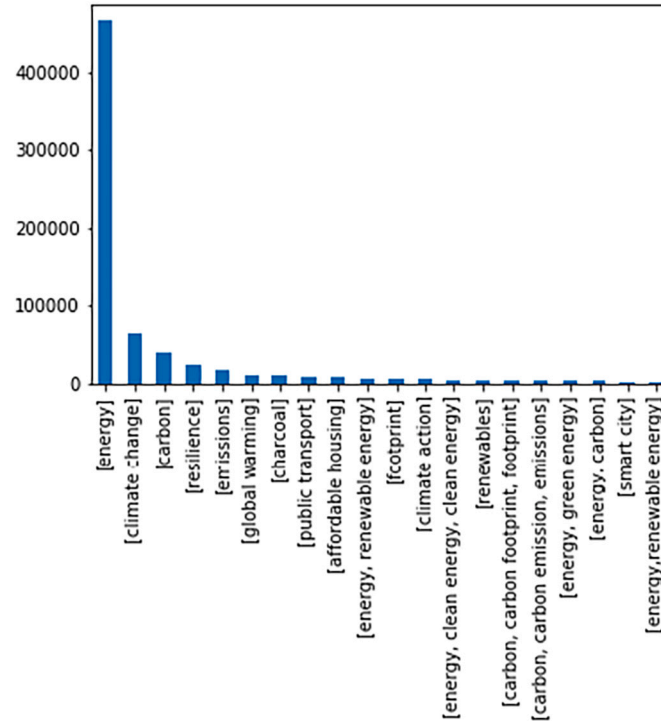


Fig. 1. Statistic over top 20 keywords after pre-processing.

Table 2
Features in the data downloaded from Twitter API.

Feature	Example
week of the year	12
day of the year	103
month	6
Text in tweet	I love Twitter
Hashtag	YOLO
User location	Washington DC

tokenized tweet. In the standard API, it is possible to pull 500,000 tweets each month and pull 50 requests each 15 minutes and in the research account, it can pull 10,000,000 tweets each month.

2.3. Data statistics

From the Twitter API a lot of features are imported. The features that were used are presented in Table 2. The data was collected between weeks 14 and 23, 2021. In total, for all the SDGs, 3,000,000 tweets were collected. That includes retweets, original tweets that mention one of the keywords, having a URL that contains it, or using a hashtag that matches one of the keywords.

Since the data was collected in real time and due to the time limit of the work, the data is unlabeled. To evaluate the different models a small part of the data was randomly selected and labeled. It was labeled as ‘negative’ or ‘positive’ (0 or 1) to evaluate the BERT-based model and negative, neutral, or positive (0, 0.5, 1) to evaluate the lexicon-based models. Due to time constraints majority vote for the labels was not possible.

2.4. Pre-processing of the data

Before using the different models the data needs to be pre-processed so the model easily can read it. The amount of pre-processing varies depending on the models, general deep learning models require much less than lexicon based since all information could be important for the

Count of Tweet by keyword

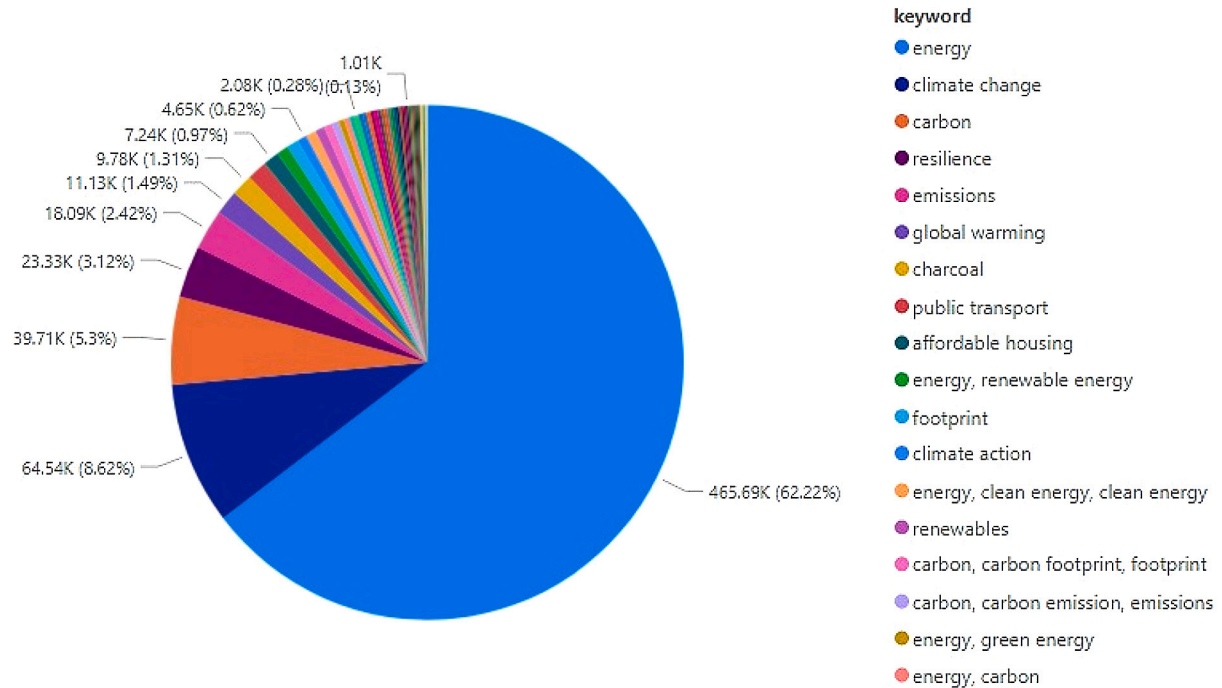


Fig. 2. Pie chart over the percentage of tweets including the different keywords in the dataset.

Table 3
Features in the training data.

Feature	Example
Target, label	0 or 4
id of the tweet	2087
Date	Sat May 16 23:58:44 UTC 2009
flag	query or NO QUERY
User	robotickilldozr
Text	Darth Vader is the best

learning process. There are a lot of models that are accessible through python packages that include pre-processing. In all the data, regardless of the model, the following pre-processing steps were made:

1. Links, HTML and HTTPURL were removed
2. Videos were removed
3. The @username was removed
4. Trailing whitespace was removed

The API collects tweets with links that contain the keyword, since the pre-processing removes the links this resulted in some tweets without the keywords. One example is Ha Haha HahahahAHAAHAAAAAAAAAAAAAA ! !

These tweets were sorted away since they did not bring any information about the sentiment of the keywords. Duplicates, and retweets, were sorted away as well. Only original tweets were saved for analysis. The total amount of data that was used in the end for testing the models was 834,558 tweets out of the initial 3,000,000.

2.5. Training data

The data set used for training the models consist of a total of 1.2 million labeled tweets and is taken from the Kaggle challenge “Sentiment140 data set with 1.6 million tweets”, for more information of the dataset see Twitter Sentiment Classification with Distant Supervision by Go et al. [30] [31]. The data was The data set is called Sentiment140 and are labeled with three different labels (0 = negative and 4 = positive). To create a binary classification model the tweets from the positive and negative class was used. We used 55,000 randomly selected samples for training. The training data only contains 53 tweets that contain any of the keywords. The training was set to 80% of the data, and test and validation were 10% each (Table 3).

3. Methods

In this section, the employed methods are presented, including the lexicon-based models VADER and TextBlob, as well as the transfer-learning method based on BERT and logistic regression.

Table 4
Lexicon scores for VADER.

Type	Score range
Positive sentiment	Compound score > 0.05
Neutral sentiment	-0.05 < Compound score < 0.05
Negative sentiment	Compound score < -0.05

Table 5
Lexicon scores for TextBlob.

Type	Score range
Polarity: negative vs. positive	(-1.0 → +1.0)
Subjectivity: objective vs. subjective	(+0.0 → +1.0)
Intensity: modifies next word?	(x0.5 → x2.0)

3.1. VADER

VADER is a lexicon-based model presented in 2004 by Hutto and Gillbert for sentiment analysis of text in micro-blog contexts, like Twitter. The model is a gold standard which means that all the classification for the lexicon is done by humans. [32] According to Hutto and Gillberts research VADER outperforms individual human rates and generalizes more favorably across contexts than any of our benchmarks [33]. For an overview of the method and process used to develop VADER see the original paper Hutto, C. & Gilbert, E. Vader: A parsimonious rule-based model for sentiment analysis of social media text (2015). [33].

VADER is available as a python package, `vaderSentiment` and is easy to use which makes it a popular option for sentiment analysis. VADER takes into account uppercase vs lowercase letters, emojis, punctuation, smileys, and slangs. VADER gives its sentiment score in four categories compound, pos, neg, and neu. The compound score is computed by summing the valence scores (goodness) of each word in the lexicon, adjusted according to the rules, and then normalized to be between -1 (most extreme negative) and +1 (most extreme positive). The recommended and used categorization for the compound score is presented in Table 4. Pos, Neg, and Neu are ratios for proportions of text that fall in each category, positive, negative, and neutral [34], respectively.

Here are some examples of application of VADER to various sentences taken from [34]:

```
1) "VADER is smart, handsome, and funny."
{'pos': 0.746, 'compound': 0.8316, 'neu': 0.254, 'neg': 0.0}

2) "VADER is VERY SMART, handsome, and FUNNY!!!"
{'pos': 0.767, 'compound': 0.9342, 'neu': 0.233, 'neg': 0.0}

3) "The book was only kind of good"
{'pos': 0.303, 'compound': 0.3832, 'neu': 0.697, 'neg': 0.0}

4) "Today SUX!"
{'pos': 0.0, 'compound': -0.5461, 'neu': 0.221, 'neg': 0.779}

5) "Make sure you :) or :D today!"
{'pos': 0.706, 'compound': 0.8633, 'neu': 0.294, 'neg': 0.0}
```

In examples 1 and 2, it is possible to see that VADER can use booster words, such as very, and that the positive sentiment increases in this case. Example 3 shows that qualified positive sentences can also be handled, example 4 and 5 show how VADER handle slang and smileys, respectively.

3.2. TextBlob

TextBlob is an open-source python package for NLP analysis that is partly based on the WordNet lexical database for English, it provides part-of-speech tagging, noun phrase extraction, sentiment analysis. The sentiment analysis scoring is divided into two different scores:

- **Polarity**
Polarity tells us the sentiment of the sentence, and its scale is between -1 (very negative) and 1 (very positive).
- **Subjectivity**
Subjectivity tells us how much the sentence is influence by the writers feeling, taste and opinions, and its scale is between 0 (very objective) and 1 (very subjective).

Each word in TextBlob's lexicon has a score of polarity, subjectivity, and intensity. The scores for each category are presented in Table 5.

3.3. Bidirectional encoder representations from transformers (BERT) model

Bidirectional Encoder Representations from Transformers (BERT) model was developed by Devlin et al. 2019 at Google [35]. BERT is based on transformers and is also a non-directional model, i.e. it takes whole sentences as input instead of sequential ordering similar to recurrent neural

Table 6
Lexicon accuracy scores for the three models.

Metrics	TextBlob	VADER	BERT
Weighted average F1-score	0.47	0.54	0.69
Weighted average precision	0.53	0.54	0.69
Accuracy	0.46	0.56	0.69

networks (RNNs) and LSTMs does. Furthermore, BERT does not use the whole transformers architecture, it only uses the encoder. BERT has two training methods: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP).

Masked Language Modeling (MLM)

To enable the training of a bidirectional model some percentage (for example 15%) of the input tokens are masked. This prevents the words to see themselves and therefore prevents the predictions to be trivial. These masked words are then the target for prediction and the target is to classify them with respect to the visible words. The final vectors that correspond to the hidden words are then fed to a standard softmax output over the vocabulary.

Next Sentence Prediction (NSP)

To understand the relationship between two sentences the model receives two sentences at the time, A and B . For 50% of the time sentence B is a random sentence and the other times it is the next sentence that follows A . Each sentence is pre-processed before it is fed into the model. The first token in every sequence is the special classification token [CLS] and the last is the special token [SEP] that separates the sentences. The input sequence is sent to a transformer encoder that predicts whether the sentence A is connected to B . The last layer is a softmax function that calculates the probabilities of the prediction. For more information see the original paper [35].

The pre-trained model that was used for creating the embeddings was trained from the whole English Wikipedia. In the present work, the input tweets were tokenized with a specialized tokenizer for BERT. The output from this BERT model (applied with HuggingFaces, a python package for transformers) was then passed to a logistic regression layer. [36] The regression layer was created with scikit-learn and trained on the data presented in the section “Training data” [37]. The sentiment classification for this model was made in two steps. First, the pre-trained BERT model gave the embedding of the tweet, then it was passed through the trained regression model.

The regression model was trained using grid search of the solvers ‘liblinear’, ‘sag’, ‘saga’ and penalty ‘l1’, ‘l2’, ‘none’. Grid search helps tuning the hyper-parameters by testing and validating all the given combinations.

In the two rule-based (dictionary-based) models, VADER and TextBlob, the sentiment is a continuous value between 1 and -1, as mentioned in the previous sections. With this, it is possible to classify the tweet with the classes positive (1), neutral (0), and negative (-1). For the BERT-based method only binary training data was available, positive (1) and negative (0). This difference in classes will make the models harder to compare and will affect the results of the classifications.

3.4. Model evaluation

The best regression model for the output from BERT was given by grid search, and resulted in using the solver ‘liblinear’ and the penalty ‘l1’. The best model got an accuracy of 79,13%. To test the models 250 tweets were annotated with two (positive and negative) and three (positive, neutral, and negative) labels. The data with three labels were used to test VADER and TextBlob and the binary data was used for BERT.

Support : The number of samples of actual occurrences of the classes in the test dataset.

$$\text{Recall} : \frac{TP}{TP + FN}$$

$$\text{Precision} : \frac{TP}{TP + FP}$$

$$\text{F1-score} : \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \quad (1)$$

$$\text{Accuracy} : \frac{TP + TN}{TP + FP + TN + FN}$$

$$\text{Weighted average} : \frac{1}{\sum_{l \in L} |y_l|} \sum_{l \in L} |y_l| P(y_l, \hat{y}_l)$$

Where TP is true positive, FP false negative, TN true negative, and FN false negative. In the expression for the weighted average y is the set of true label pairs, $\hat{y}$ the set of predicted label pairs, L the set of labels, y_l the subset set of label l (similarly to $\hat{y}_l$ and $P(A, B) = \frac{|A \cap B|}{|B|}$). The confusion matrix for each model is shown in Fig. 3 where the x-axis is the predicted label and the y-axis is the true label and the result for each model is presented in Table 6, the results are calculated with the equations presented in equation (1). The confusion matrix shows the number of tweets (of the 248 annotated tweets) that were correctly and wrongly labeled. It is possible to see that BERT has the highest amount of tweets that were correctly labeled as positive.

4. Results

The results will be divided into two parts: first an analysis of the sentiments from the different methods and then the accuracy of the different models. The pie chart depicted in Fig. 2 shows the percentage of the keywords, and the list on the right is the most common keywords in the collected data set. In this section, the transfer-learning model based on BERT and regression will be referred to as the BERT model or only BERT.

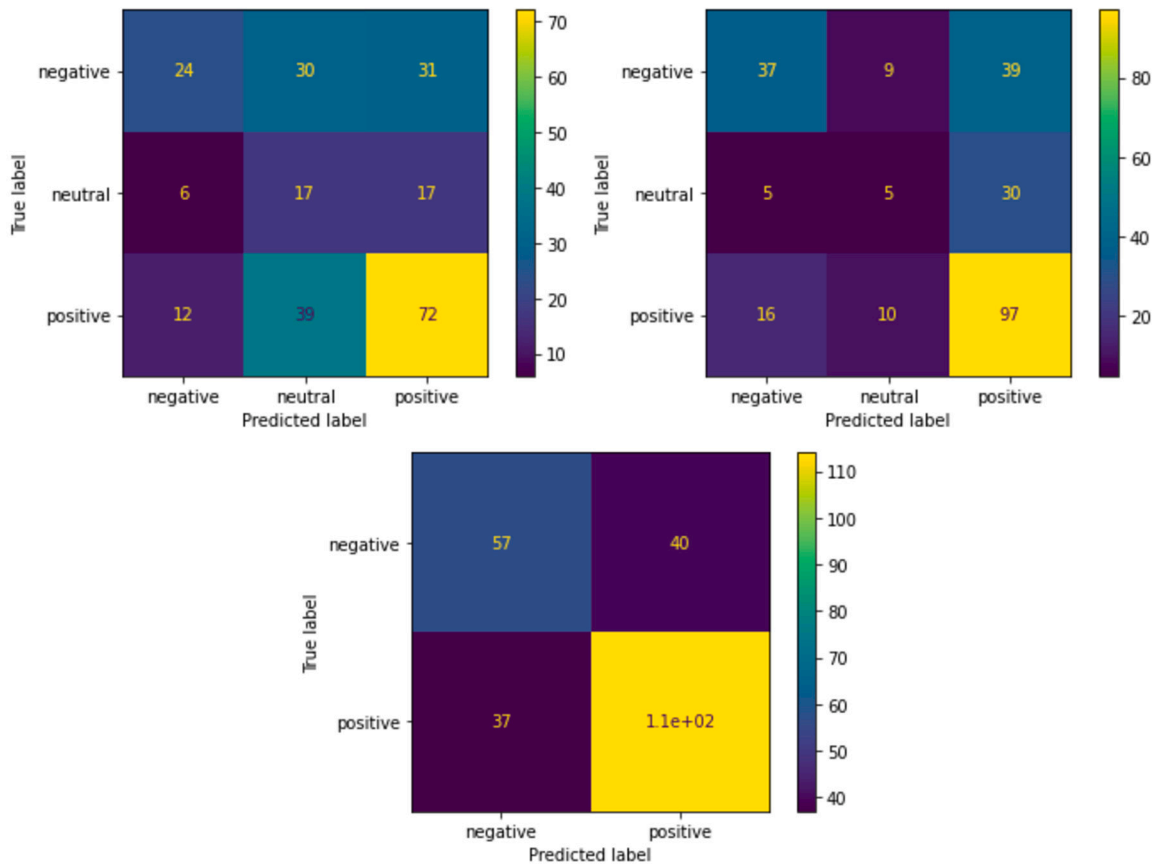


Fig. 3. Confusion matrix for the result of TextBlob (top left), VADER (top right) and BERT (bottom).

4.1. Sentiment analysis

As mentioned above, the various methods have different amounts of labels. To compare them between each other the results from the sentiment score from the VADER and TextBlob method were normalized to a scale of 0 to 1 to match the range of the BERT score. It was normalized with min-max scaling, see equation (2). The normalized data were then used to compare the models directly against each other as follows (note that X denotes the value to scale within its range):

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}. \quad (2)$$

To get an overview of the data and the difference between the two rule-based models, VADER and TextBlob, see Fig. 4 which shows the percentage stack bar for the two models. Blue color represents the positive class, orange shows the negative, and green indicates the neutral. This figure shows the overall trends in the data.

The words with the highest percentage of positive and negative tweets can be seen in Table 7.

The trend can be visualized with histograms over the amount of positive, negative, and neutral classification by VADER and TextBlob, shown in Fig. 5 and positive and negative classification by BERT in Fig. 6.

In our case, we do not have any certain event to compare against, but it is still interesting to see how the sentiment changed during the period in which the data was collected. Fig. 7 shows the average normalized sentiment score for each week. It is possible to see that the three models follow the same trends with a peak around weeks 21 and 22.

The results for the grouped keywords within each SDG are shown in Fig. 8, 9 and 10.

The sentiments for the different goals are very similar, as can be seen in the Table 8 that shows the ratio between the amount of positive tweets and amount of negative tweets.

5. Discussion and conclusions

One of the most challenging parts when working with text data is the pre-processing of the data. The human written language has many different varieties in forms of slang, dialects, local words, spelling and much more. This has been shown in the results. The best model was the transfer learning model with BERT and logistic regression, it got 0.69 in accuracy as seen in Table 6. This is quite good considering the training score for the model was 0.7913 and the low amount of test data, 248. The dictionary-based models were not as good as the BERT-based model, with an accuracy of only 0.56 (VADER) and 0.46 (TextBlob) they were quite bad. The result is not that surprising considering the complexity that the pre-trained BERT model has. It is trained on large corpora and by that, it can see more complex structures in the text. To improve the BERT-based model BERTweet (a pre-trained BERT model trained on tweets) could be used, since it is trained on Twitter data it would improve the model's capacity to understand the short text that is typical for Twitter.

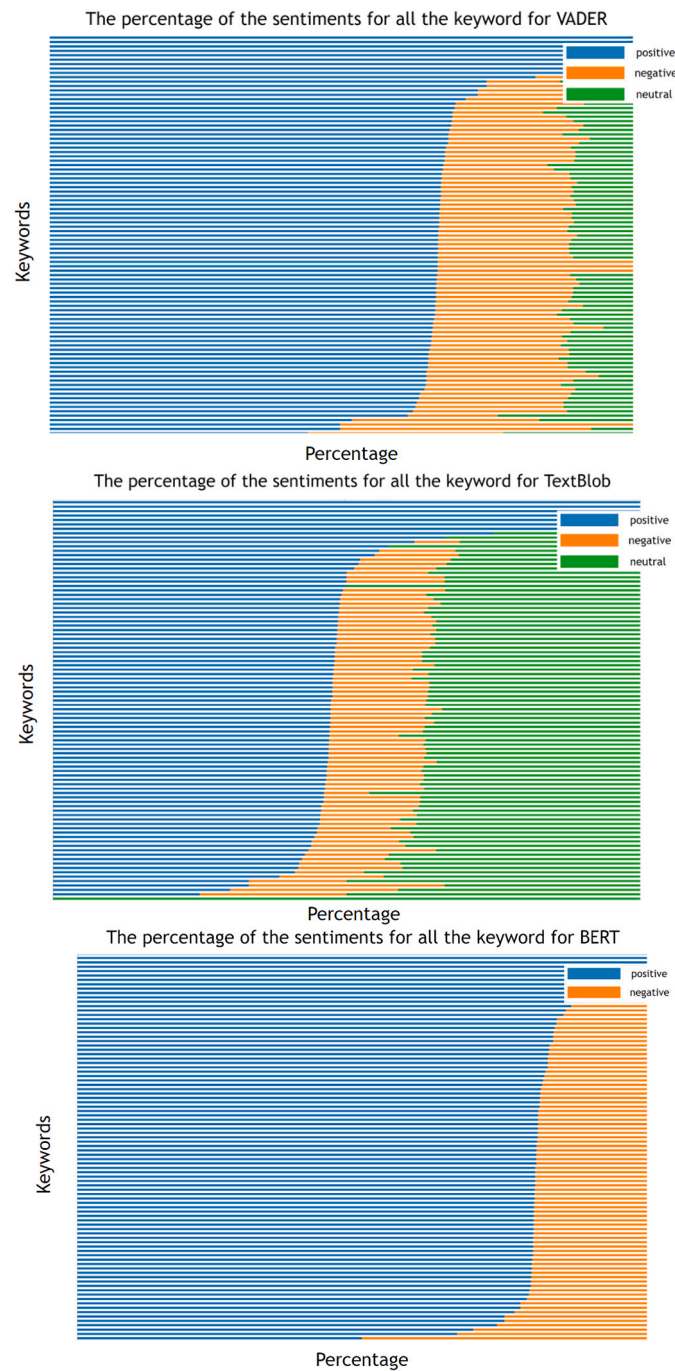


Fig. 4. Percentage stack bar for the sentiment scores of VADER (top) and TextBlob (middle) and BERT (bottom). Blue color represents the positive sentiment, orange shows the negative sentiment and green indicates the neutral sentiment.

A large challenge with the rule-based models when using collected data based on keywords is that the keyword itself has a value. Energy for example has a sentiment score of 0.27 from VADER. This means that every tweet that contains energy will have a positive addition due to the keyword itself. The model will then have a bias because of this. With the pre-trained models, it is, therefore, better to use a deep-learning approach for this type of analysis. VADER performs better than TextBlob, it can be because VADER is created for Twitter data.

The validation, in this case, was based on self-annotated data, and therefore the human factor plays a large role, and depending on the annotator the result differed. To make the validation more robust multiple annotators should annotate the same data. This was not possible in this project due to time limitations.

The low amount of occurrences of the keywords in the training data, only 53 tweets of 55000 contained any of the keywords, could explain the accuracy of the models. With a training set that contains more of the keyword, it would have a chance to improve the results.

The selection of the keywords also plays a large role in the results, in this case, the chosen words are not the most common words to use in everyday language. For example waste management, biomass and urbanization are not that commonly used in social media contexts, especially on Twitter which is based on short posts. It was possible to see this when looking at the most common keywords, seen in Fig. 1.

Table 7
The words with the most negative percentage (left) and the words with the most positive (right).

Negative	Positive
climate impact	city air quality
climate resilience	climate-related hazards
climate policy	biosequestration
gas emission	biogasoline
biofuels	sustainable rural development
megacities	woodfuels
charcoal	sustainable footprint
photovoltaic	sustainable urbanization
sustainable transport	climate national
climate mitigation	climate international
climate refugees	city pollution
mega cities	resilient buildings
photovoltaics	climate planning
human resource planning	climate hazards
population surveillance	climate hazard
household energy	sustainable housing
climate variability	modern energy
national development planning	affordable energy
household resilience	sustainable construction

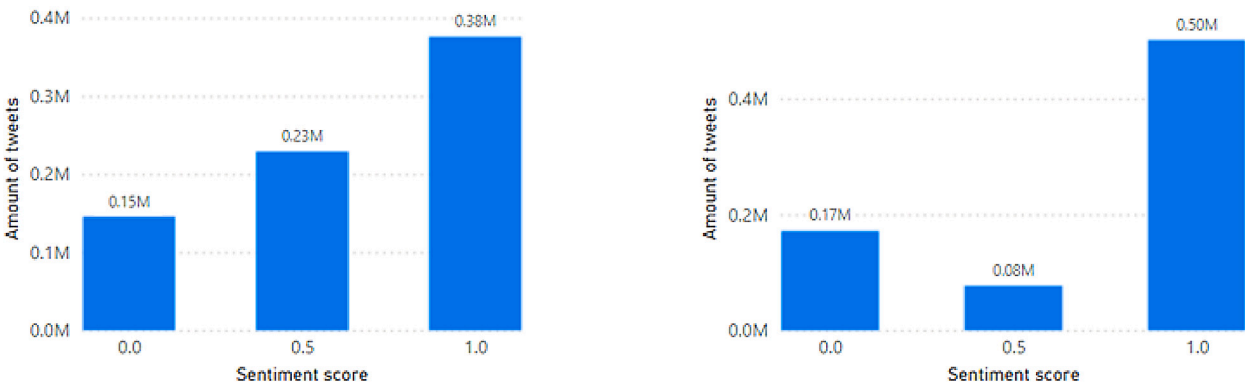


Fig. 5. Count of each sentiment classified by TextBlob (left) and VADER (right).

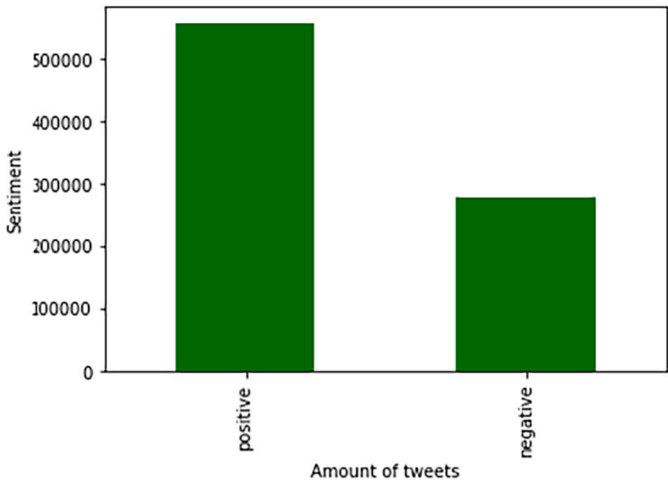


Fig. 6. Count of each sentiment classified by BERT.

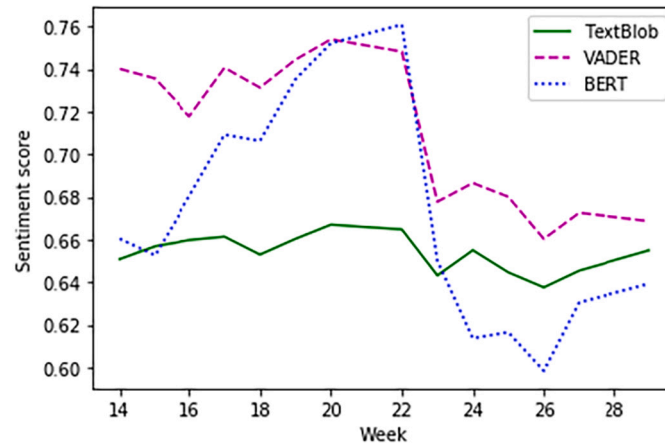


Fig. 7. Average normalized sentiment scores for each week.

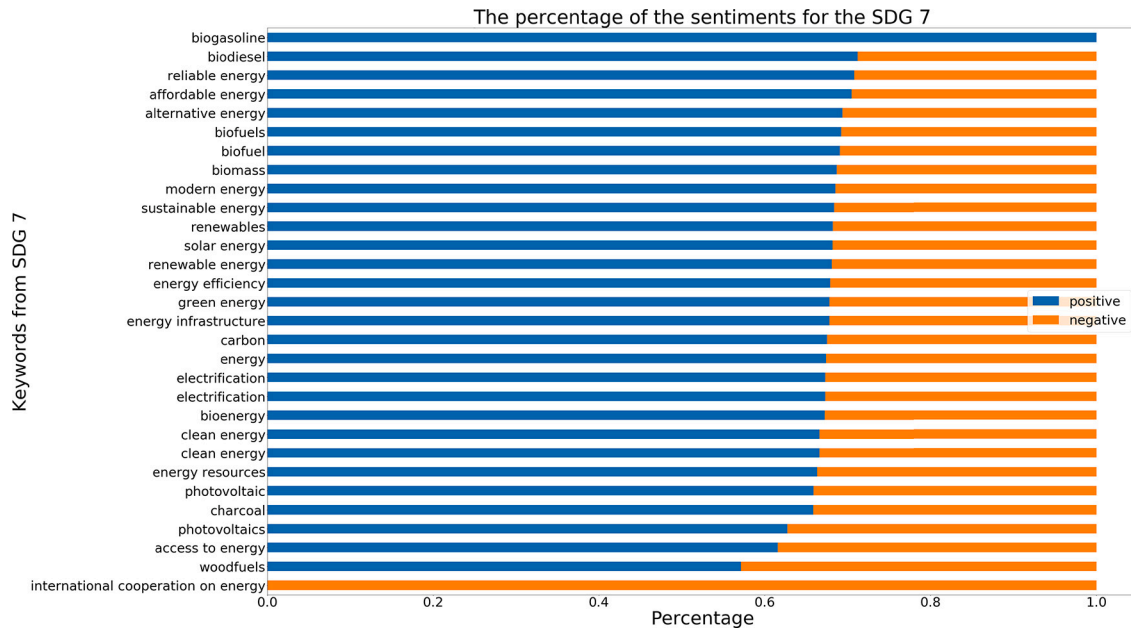


Fig. 8. Stack bar for SDG 7.

Table 8
Ratio between amount of positive and negative tweets for each SDG.

SDG	Ratio
7	0.4849
11	0.47748
13	0.4952

Energy has by far the largest representation in the data because of the versatility of the word. Energy can refer to things like “I like green energy” but also to “I have so much energy today”. This is a good example of a poorly chosen keyword. The words should preferably be more precise and used in everyday language. To tackle the challenge with generic keywords, such as Energy, more advanced data pre-processing techniques are needed to sort the data to make sure that the content of the tweet is within the right domain, in this case, climate. One may try to use additional NLP techniques to determine this, rule-based or more advanced AI-based methods. The collection of a climate-related dataset that are based on more than keywords would be needed to avoid this problem.

The use of other additional social media platforms would give a more clear picture of the opinion related to the SDG goals. The hope was to get an idea about the opinion and engagement about a topic related to the UN’s SDGs. This was hard due to the high amount of noise in the data, both from the keywords and from the lack of filters in the Twitter API. To make the data more robust and suitable for the analysis additional filters should be added to the API. Only 27,8% of the original data was used, this is because of two reasons. First, due to the filtering from Twitter API, some tweets contain the keywords in the link they shared and slip through, the second reason is the high amount of duplicates. Every re-tweet of an existing tweet in the data set will be a duplicate. These could be seen as enforcement of the opinion in the original tweet and used in the analysis.

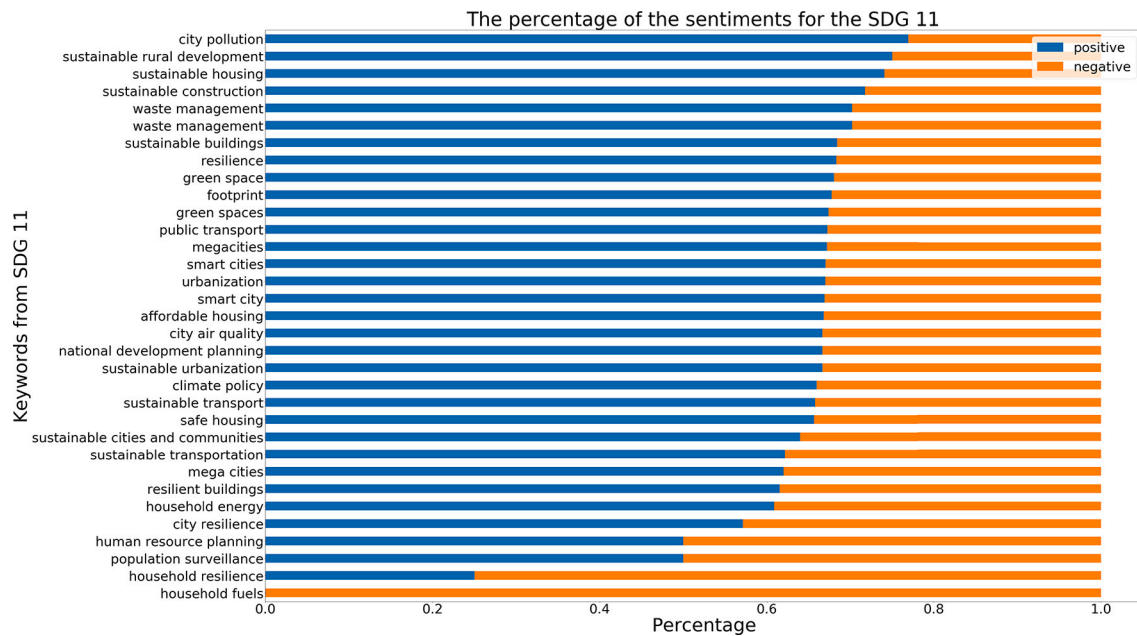


Fig. 9. Stack bar for SDG 11.

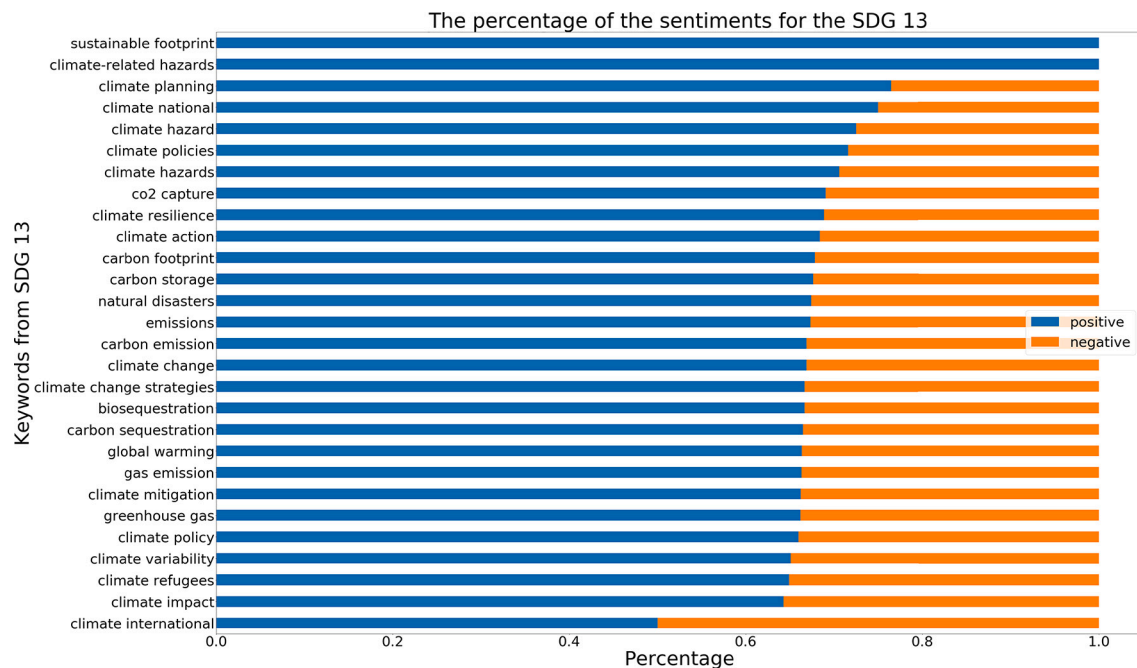


Fig. 10. Stack bar for SDG 13.

This would lead to more data but it can also be misleading since we do not know why the user retweeted that tweet, it could be because they shared the opinion or the opposite.

Another downside of the Twitter API is that each collected tweet does not get linked to its corresponding keyword. This means that every tweet that is collected is also scanned for the chosen keywords. This makes the chance of missing a keyword much bigger since it only scans for the given conjugation of that word. This can be fixed with NLP techniques but it adds an unnecessary obstacle.

It is hard to draw any definitive conclusions about the overall sentiment surrounding the chosen SDGs. Nonetheless, from the present data and analysis, we can determine that the sentiment is mostly positive, even in tweets associated with “bad” keywords (e.g. climate hazards, pollution). This proves the complexity and challenge of sentiment analysis. This unexpected result could be because of sarcasm, lack of data from that specific word, or contradictory meanings. For example, the sentence ‘I love our cities initiative against city pollution has a positive sentiment score but is good for the climate. So even if the result may indicate that people are talking positively about city pollution it is the opposite, they talk positively about efforts against it.

It is possible to see changes in the sentiments over time, in this case, the number of tweets varied each week which adds uncertainty in the analysis because this could affect the total sentiment. To draw conclusions from the time series it needs to be compared to a specific event, then it would be possible to see the changes before, during, and after.

By using the sentiment from keywords to compare the different SDGs it could be seen that the ratio between positive and negative sentiments was almost identical. To be able to measure how well sentiment would work as a proxy for SDG achievements, a larger study with SDG from different areas, like health and environment, is needed to fully understand the potential. At the same time, it is interesting that the result for the different groups of keywords is almost identical. They consist of different words within the same topic. It is possible to draw the conclusion that when it comes to climate-related issues the public speaks positively. What that truly means requires more research but we can see a connection between the SDGs.

6. Conclusions

In conclusion, BERT-based models outperform the rule-based models VADER and TextBlob when it comes to precision, recall, and F1-score. Rule-based models add a bias in the classifications when using data collected with keywords due to the sentiment score of the keyword itself. More complex deep-learning models like BERT do not result in the same bias since it is based on context rather than rules. By adding a more comprehensive filter to Twitter API the data could be more suitable for the analysis. This could decrease the amount of noise and outliers in the dataset. In all of the different SDGs, the majority of tweets were positive and all had a similar pattern in the distribution. Most of the keywords had between 60-70% positive tweets. The similarities between the different SDGs and the fact that they cover the same topic could mean that sentiment has the potential to be a proxy for SDG achievements. To fully understand it a similar study should be made with keywords related to SDGs related to other topics, such as SDG 3 on health. The overall sentiment on Twitter about the keywords is positive, and the same yields for the sentiments for each SDG.

CRediT authorship contribution statement

ER performed the analysis, with the help of CT, FM and HE. The supervision was carried out by DPE, FFN and RV. ER wrote the article with input from all the authors.

All authors contributed equally to this project.

Declaration of competing interest

The authors declare that they have no competing interests.

Data availability

All the data and codes used to produce the present study will be made available open access upon publication of the manuscript.

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Appendix A

A.1. Keywords for SDG 7

Affordable energy, energy, reliable energy, modern energy, access to energy, electrification, clean energy, renewable energy, energy efficiency, renewables, energy infrastructure, fossil-fuel technology, clean energy, international cooperation on energy, alternative energy, energy resources, solar energy, photovoltaic, photovoltaics, electrification, bioenergy, biofuel, biofuels, biodiesel, biogasoline, carbon, charcoal, green energy, biomass, woodfuels, sustainable energy, sustainable energy investments, energy developing countries, energy land-locked countries, energy least developed countries.

A.2. Keywords for SDG 11

Safe housing, affordable housing, upgrade slums, sustainable transport, sustainable transportation, public transport, city air quality, waste management, sustainable cities and communities, sustainable housing, urbanization, urban environmental impact, climate policy, footprint, green space, green spaces, household energy, household resilience, resilience, household fuels, human activity environment, human resource planning, human sensors, mega cities, mega cities, neighborhood sustainability assessment, population surveillance, smart cities, smart city, environmental inequality, waste management, national development planning, regional development planning, sustainable rural development, city pollution, city mitigation, city adaptation climate change, city resilience, resilient buildings, sustainable buildings, buildings local materials, sustainable urbanization, sustainable construction.

A.3. Keywords for SDG 13

Climate resilience, natural disasters, climate hazard, climate hazards, climate impact, climate mitigation, climate policy, climate policies, climate action, climate change strategies, climate planning, climate national, climate international, climate change, climate-related hazards, COP, COP26, COP25, greenhouse gas, gas emission, social footprint, sustainable footprint, biosequestration, carbon footprint, carbon emission, carbon economy, carbon storage, carbon sequestration, global warming, emissions, environmental change, co2 capture, climate variability, climate refugees, climate biodiversity, climate desertification, climate change developed countries, Paris Agreement, climate change marginalized communities, UNFCCC, negotiation global response climate change, global warming commitment.

References

- [1] K. Granville, Facebook and Cambridge Analytica: What You Need to Know as Fallout Widens, 2018.
- [2] R. Vinuesa, B. Sirmacek, Interpretable Deep-Learning Models to Help Achieve the Sustainable Development Goals, 2021.
- [3] The sustainable development agenda - United Nations sustainable development.
- [4] B. Pang, L. Lee, A sentimental education: sentiment analysis using subjectivity summarization based on minimum cuts, in: Proceedings of the 42nd Annual Meeting on Association for Computational Linguistics, ACL '04, Association for Computational Linguistics, USA, 2004, 271–es.
- [5] P.D. Turney, Thumbs up or thumbs down? Semantic orientation applied to unsupervised classification of reviews, in: Proceedings of the 40th Annual Meeting on Association for Computational Linguistics, ACL '02, Association for Computational Linguistics, USA, 2002, pp. 417–424.
- [6] K. Dave, S. Lawrence, D. Pennock, Mining the peanut gallery: opinion extraction and semantic classification of product reviews, *Min. Peanut Gallery: Opin. Extr. Semantic Classif. Prod. Rev.* (2003), <https://doi.org/10.1145/775152.775226> 775152.
- [7] O. Appel, F. Chiclana, J. Carter, H. Fujita, A hybrid approach to the sentiment analysis problem at the sentence level, in: New Avenues in Knowledge Bases for Natural Language Processing, *Knowl.-Based Syst.* 108 (2016) 110–124, <https://doi.org/10.1016/j.knosys.2016.05.040>.
- [8] T. Wilson, J. Wiebe, P. Hoffmann, Recognizing contextual polarity in phrase-level sentiment analysis, in: Proceedings of the Conference on Human Language Technology and Empirical Methods in Natural Language Processing, HLT '05, Association for Computational Linguistics, USA, 2005, pp. 347–354.
- [9] R. Moraes, J. Valiati, W. Gavião Neto, Document-level sentiment classification: an empirical comparison between svm and ann, *Expert Syst. Appl.* 40 (2013) 621–633, <https://doi.org/10.1016/j.eswa.2012.07.059>.
- [10] R. Johnson, T. Zhang, Semi-supervised convolutional neural networks for text categorization via region embedding, *Adv. Neural Inf. Process. Syst.* 28 (2015) 919–927.
- [11] C. Guggilla, T. Miller, I. Gurevych, Cnn- and lstm-based claim classification in online user comments, in: COLING, 2016.
- [12] T. Mikolov, K. Chen, G. Corrado, J. Dean, Efficient estimation of word representations in vector space, in: Proceedings of Workshop at ICLR 2013, 2013.
- [13] S. Ruder, P. Ghaffari, J. Breslin, A Hierarchical Model of Reviews for Aspect-Based Sentiment Analysis, 2016.
- [14] A. Vaswani, et al., Attention is all you need, arXiv:1706.03762, 2017.
- [15] K. Pipalia, R. Bhadja, M. Shukla, Comparative analysis of different transformer based architectures used in sentiment analysis, in: 2020 9th International Conference System Modeling and Advancement in Research Trends (SMART), 2020, pp. 411–415.
- [16] N. S. D. Paulraj, P. Ezhumalai, M. Prakash, A deep learning modified neural network (dlmnn) based proficient sentiment analysis technique on Twitter data, *J. Exp. Theor. Artif. Intell.* (0 2022) 1–20, <https://doi.org/10.1080/0952813X.2022.2093405>.
- [17] D. Zimbra, A. Abbasi, D. Zeng, The state-of-the-art in Twitter sentiment analysis: a review and benchmark evaluation, *ACM Trans. Manag. Inf. Syst.* 9 (2) (2018) 1–29, <https://doi.org/10.1145/3185045>.
- [18] M.L. Loureiro, M. Alló, Sensing climate change and energy issues: sentiment and emotion analysis with social media in the U.K. and Spain, *Energy Policy* 143 (2020) 111490, <https://doi.org/10.1016/j.enpol.2020.111490>.
- [19] A. Kirilenko, S. Stepchenkova, Public microblogging on climate change: one year of Twitter worldwide, *Glob. Environ. Change* 26 (2014) 171–182, <https://doi.org/10.1016/j.gloenvcha.2014.02.008>.
- [20] D. Maynard, K. Bontcheva, Understanding climate change tweets: an open source toolkit for social media analysis, <https://doi.org/10.2991/ict4s-env-15.2015.28>, 2015.
- [21] A. Reyes-Menendez, J.R. Saura, C. Alvarez-Alonso, Understanding worldenvironmentday user opinions in Twitter: a topic-based sentiment analysis approach, *Int. J. Environ. Res. Public Health* 15 (2018), <https://doi.org/10.3390/ijerph15112537>.
- [22] R. Vinuesa, et al., The role of artificial intelligence in achieving the sustainable development goals, *Nat. Commun.* 11 (2020).
- [23] S. Gupta, et al., Assessing whether artificial intelligence is an enabler or an inhibitor of sustainability at indicator level, *Transp. Eng.* 4 (2021) 100064, <https://doi.org/10.1016/j.treng.2021.100064>.
- [24] S. MacFeely, The big (data) bang: opportunities and challenges for compiling sdg indicators, *Glob. Policy* 10 (2019) 121–133, <https://doi.org/10.1111/1758-5899.12595>, <https://onlinelibrary.wiley.com/doi/pdf/10.1111/1758-5899.12595>.
- [25] A. Sánchez-Roncero, et al., The sustainable development goals and aerospace engineering: a critical note through artificial intelligence, *Results Eng.* 17 (2023) 100940, <https://doi.org/10.1016/j.rineng.2023.100940>.
- [26] H.H. Goh, R. Vinuesa, Regulating artificial-intelligence applications to achieve the sustainable development goals, *Discov. Sustain.* 2 (2021) 52.
- [27] K. Baum, et al., From fear to action: AI governance and opportunities for all, *Front. Comput. Sci.* 5 (2023).
- [28] F. Larosa, et al., Halting generative AI advancements may slow down progress in climate research, *Nat. Clim. Change* 13 (2023) 497–499.
- [29] J. Roesslein, Tweepy: Twitter for python!, <https://github.com/tweepy/tweepy>, 2020.
- [30] KazAnova, Sentiment140 Dataset with 1.6 Million Tweets, 2017.
- [31] A. Go, R. Bhayani, L. Huang, Twitter sentiment classification using distant supervision, *Processing* 150 (2009).
- [32] C.J. Hutto, E. Gilbert, Vader: a parsimonious rule-based model for sentiment analysis of social media text, in: ICWSM, 2014.
- [33] C. Hutto, E. Gilbert, Vader: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text, 2015.
- [34] Cjhutto, Cjhutto/vadersentiment: Vader sentiment analysis. Vader (valence aware dictionary and sentiment reasoner) is a lexicon and rule-based sentiment analysis tool that is specifically attuned to sentiments expressed in social media, and works well on texts from other domains.
- [35] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: pre-training of deep bidirectional transformers for language understanding, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), Association for Computational Linguistics, Minneapolis, Minnesota, 2019.
- [36] Hugging face – the ai community building the future.
- [37] F. Pedregosa, et al., Scikit-learn: machine learning in python, *J. Mach. Learn. Res.* 12 (2011) 2825–2830.